

Patient Profile

Saubhik Bhaumik, 27-year-old male

Summary of Results

Liver function tests demonstrate values within established reference ranges, indicative of normal hepatic function.

ML Result: Normal— the predictive model assesses the profile as within normal limits.

Detailed Test Explanations

Serum Bilirubin (Total)

: Measures total bilirubin, a byproduct of red blood cell breakdown processed by the liver; elevated levels may suggest hemolysis, liver dysfunction, or biliary obstruction. Patient result: 0.9 mg/dL (reference: 0.2-1.2 mg/dL) – normal.

Serum Bilirubin (Direct)

: Quantifies conjugated bilirubin excreted via bile; increases can indicate hepatocellular or obstructive liver disease. Patient result: 0.2 mg/dL (reference: 0-0.3 mg/dL) – normal.

Serum Bilirubin (Indirect)

: Reflects unconjugated bilirubin; elevations often relate to hemolysis or impaired hepatic conjugation. Patient result: 0.70 mg/dL (reference: 0.2-1.0 mg/dL) – normal.

SGPT (ALT)

: Alanine aminotransferase, a liver enzyme highly specific for hepatocellular injury; elevations signal acute liver damage from toxins, ischemia, or viral hepatitis. Patient result: 36 U/L (reference: 13-40 U/L) – normal.

SGOT (AST)

: Aspartate aminotransferase, present in liver, heart, and muscle; rises indicate tissue damage, though less liver-specific than ALT. Patient result: 32 U/L (reference: 0-37 U/L) – normal.

Serum Alkaline Phosphatase

: Enzyme from liver, bone, and intestine; increased levels suggest cholestasis, bone disease, or biliary obstruction. Patient result: Not assessed.

Serum Protein

: Total protein reflecting nutritional status and liver synthetic function; hypoalbuminemia or hypoproteinemia may indicate chronic liver disease or malnutrition. Patient result: 7.2 g/dL (reference: 6.4-8.3 g/dL) – normal.

Serum Albumin

: Primary protein synthesized by the liver, essential for oncotic pressure and transport; low levels denote chronic liver disease, nephrotic syndrome, or inflammation. Patient result: 4.7 g/dL (reference: 3.5-5.2 g/dL) – normal.

Globulin

: Immunoglobulins and other proteins; derangements may reflect immune dysregulation or chronic inflammation. Patient result: 2.50 g/dL (reference: 1.8-3.6 g/dL) – normal.

A/G Ratio

: Albumin-to-globulin ratio assessing synthetic balance; imbalance suggests chronic liver disease or protein loss. Patient result: 1.88 (reference: 1.1-2.4) – normal.

Overall Interpretation

All assessed liver function parameters fall within normal reference intervals, consistent with intact hepatocellular integrity, bilirubin metabolism, and protein synthesis. The absence of enzyme elevations or bilirubin abnormalities supports no evidence of acute or chronic liver pathology.

ML Result: Normal reinforces this profile as within physiological norms.

Recommendations

Maintain a balanced diet rich in fruits, vegetables, and lean proteins; limit alcohol intake to promote sustained liver health. Engage in regular moderate exercise and monitor weight. Routine follow-up testing in 12 months or sooner if symptoms such as fatigue, jaundice, or abdominal pain emerge.

Correlate with clinical history as needed.

Disclaimer

This report is for informational purposes and does not substitute professional medical advice, diagnosis, or treatment. Consult a healthcare provider for personalized interpretation.